

MT VERNON OUTLAND, IL, USA

WMO: 724335

Lat: 38.323N

Lon: 88.858W

Elev: 480

StdP: 14.44

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93894

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.5	(c) 10.5	(d) -4.9	(e) 4.3	(f) 7.8	(g) 0.8	(h) 5.8	(i) 13.2	(j) 24.5	(k) 52.5	(l) 21.5	(m) 51.2	(n) 3.9	(o) 280	(p) 0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.8	(c) 94.7	(d) 76.9	(e) 91.5	(f) 76.3	(g) 90.2	(h) 75.6	(i) 80.2	(j) 89.7	(k) 78.4	(l) 88.7	(m) 77.1	(n) 87.3	(o) 8.7	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	145.1	86.6	75.2	134.5	85.0	73.4	126.5	82.7	44.0	90.2	42.4	88.8	40.8	87.1	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
20.7	18.0	15.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-4.5	98.0	7.5	3.0	-9.9	100.1	-14.2	101.9	-18.4	103.6	-23.9	105.8		
			-4.6	81.7	7.6	2.4	-10.1	83.4	-14.6	84.8	-18.9	86.2	-24.4	87.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	31.3	35.6	44.7	55.7	65.0	74.0	77.1	75.8	68.5	56.2	44.1	35.5
	DBStd	18.44	11.90	11.25	10.77	9.12	8.00	6.11	5.47	5.35	7.38	9.17	9.56	10.74
	HDD50	2021	589	420	235	46	2	0	0	0	0	40	226	463
	HDD65	4806	1044	824	634	301	104	8	0	2	48	299	627	915
	CDD50	3991	10	16	70	215	467	721	840	800	555	234	50	13
	CDD65	1302	0	0	3	20	104	280	375	338	153	28	1	0
	CDH74	13308	0	0	27	188	977	2945	3921	3348	1576	314	12	0
	CDH80	5531	0	0	1	19	282	1265	1793	1468	627	75	1	0
(14) Wind	WSAvg	5.6	6.5	6.5	6.9	7.3	5.7	5.0	4.4	4.2	4.1	5.0	6.0	6.1
(15) Precipitation	PrecAvg	41.2	2.3	2.5	4.0	4.3	4.6	3.7	3.7	3.1	3.1	2.9	4.1	3.1
	PrecMax	56.4	6.9	6.4	9.4	9.4	12.9	9.7	9.3	9.0	8.4	10.0	13.1	10.2
	PrecMin	30.0	0.0	0.3	1.0	1.7	0.7	0.2	0.9	0.3	0.5	0.3	0.5	0.6
	PrecStd	6.8	1.6	1.5	2.0	1.9	2.6	2.0	2.2	1.9	1.9	2.0	2.4	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.1	69.4	78.6	82.1	90.1	96.6	99.6	98.9	94.7	86.3	75.0	66.1
		MCWB	58.9	60.2	64.1	66.4	73.5	75.1	77.9	76.8	73.3	71.6	61.5	60.3
	2%	DB	60.6	63.5	72.4	79.3	86.4	92.8	94.8	93.3	90.6	81.7	68.8	61.3
		MCWB	55.9	55.4	60.0	64.6	72.5	75.5	78.5	76.2	73.9	66.6	60.3	55.8
	5%	DB	55.0	59.0	67.6	75.3	82.8	90.5	91.4	90.7	87.8	77.2	64.2	56.7
		MCWB	50.5	52.4	57.1	62.7	70.1	75.1	77.7	76.1	72.5	64.4	56.9	51.4
	10%	DB	48.4	54.1	63.2	72.3	80.6	88.0	89.8	88.2	82.4	72.7	61.1	52.1
		MCWB	44.4	47.7	54.4	61.0	68.4	74.0	76.7	74.6	69.2	62.6	53.6	47.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.7	62.3	65.6	70.0	76.3	80.1	83.2	81.9	77.8	73.9	65.4	62.1
		MCDB	63.5	67.3	75.5	79.2	85.7	89.8	92.1	90.2	89.4	83.4	73.2	65.3
	2%	WB	56.4	58.1	62.0	67.3	74.2	78.3	81.1	79.2	76.0	70.2	61.6	57.1
		MCDB	60.2	62.3	70.8	76.1	83.6	89.0	90.0	89.6	87.3	77.2	66.7	61.3
	5%	WB	51.3	53.3	58.8	64.9	72.0	76.8	79.3	77.5	73.8	67.1	58.7	52.2
		MCDB	54.4	58.1	65.7	73.1	80.9	87.2	89.2	88.0	83.6	73.9	64.5	56.3
	10%	WB	44.6	47.3	55.1	62.4	69.6	75.0	77.6	75.9	71.6	64.1	54.4	46.8
		MCDB	48.6	53.0	62.3	70.0	77.6	84.4	87.7	85.3	79.9	71.2	60.1	50.0
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	17.5	20.1	22.1	21.3	22.0	20.8	21.9	24.3	23.5	19.9	16.2
		MCDBR	21.2	24.5	25.3	25.8	24.8	25.5	23.4	25.1	28.4	28.6	24.7	21.2
	5% WB	MCWBR	18.1	19.0	16.9	15.6	13.2	11.4	10.4	11.2	13.3	15.9	17.6	17.9
		MCDWR	19.2	21.7	22.4	21.5	21.5	21.1	20.8	21.4	24.1	22.9	21.2	19.0
(40) Clear-Sky Solar Irradiance	taub		0.311	0.324	0.363	0.399	0.434	0.437	0.465	0.464	0.408	0.358	0.320	0.306
		taud	2.458	2.405	2.361	2.260	2.222	2.232	2.163	2.153	2.302	2.438	2.488	2.480
	Ebn at Noon		276	286	284	279	270	268	259	257	266	270	270	269
		Edh at Noon	26	31	36	42	45	44	47	46	38	30	25	24
(44) All-Sky Solar Radiation	RadAvg		637	874	1187	1567	1761	2033	1971	1790	1493	1089	735	539
	RadStd		63	96	112	92	153	131	128	98	100	114	82	52

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140

Nomenclature: See separate page